

LABS

Lab to practice powershell in DC2

- Display list of enabled firewall rules for Public profile on DC2
- Display all and only ipv4 addresses on DC2
- Display the startup type of BITS service
- Change the startup type of BITS service
- Display all environmental variables on DC2
- Display computername variable on DC2

Lab to set up active directory domain services

- Log in to DC1 and add the Active Directory Domain Services role
 - Disable dns delegation
 - Use Win2016 domain level
 - Use Win2016 forest level
 - Use int.icXX.jc.crp as domain name where XX is student number (01..20)
 - Use INT as netbios domain
 - Install dns with ADDS
 - Add the Cool2Pass as DSRM password
 - Restart the computer if the script requires it

Lab to create one additional user

- Create new user “NewUser” with password “Cool2Pass” in the domain
- Make sure that the only group the user is in is “Domain users” and nothing else

Lab to join workstations to domain

- Join DC2 to the domain int.icXX.jc.crp
- Join WS1 to the domain int.icXX.jc.crp
- Join WS2 to the domain int.icXX.jc.crp
 - Use the newly created regular user NewUser to achieve it
- Join WS3 to the domain int.icXX.jc.crp

Lab to modify AD database with ADSIedit

- Using ADSIedit make sure that regular users cannot add workstations to domain

Lab to create OU

- Create OU named **Hosts** in the domain and move all workstations there

Lab to create GPO and apply wallpaper

- Create GPO to change wallpaper to wolf_wallpaper.jpg to apply it to the Hosts OU where workstations reside
- Create another GPO that also applies to the Hosts OU but changes the wallpaper of windows7 hosts to fox_wallpaper.jpg

Lab to transfer FSMO roles

- Transfer the PDC master role to DC2
- Add NewUser to Remote Desktop users group on WS2
- Shut down DC2
- Log in to WS2 with NewUser to WS2
- Change password of NewUser to Cool2Pass2

Lab on powershell, date and dns

- On DC1 display only the day of the year
- On DC1 display the installation date, patch (hotfix) ID number and the name of the user who installed the patch (hotfix)
- On DC1 display the enabled firewall rules but only the rules:
Display name, Profile, Direction and allow/deny and sort the list in alphabetical order by Display name
- On DC1 display list of network neighbours
 - Sort the list by state
 - Sort by state and display only IP address and state
- On DC1 display DNS cache
 - Sort the list by name and display only name, type and TTL

Lab on powershell and processes

- On WS2
 - Display list of running processes
 - Display list of running processes sorted in reverse alphabetical order by process name and show only the
 - Process name
 - ID
 - Virtual memory usage
 - Physical memory usage
 - Output the previous task list to a HTML file named processes.html
 - Add an additional line “Running processes” in the beginning of the HTML file and current date to the end of the HTML file

Lab on powershell and output types

- On WS2
 - Create a comma separated (CSV) file listing last 10 entries from System event log (system_csv.txt)
 - Remove the first line that begins with #
 - Create XML file of all services (services.xml)
 - Create a XML file of all services but only with Service name, Display name and status (services2.xml)
 - Create a pipe-delimited list of the most recent 20 Security event log entries (security_pipe.txt)
 - Only have the eventid, time and message in output file (security_pipe2.txt)
 - Import data from pipe-delimited file (security_pipe.txt) into the variable “pipetable”

Lab on creating AD users

- Create 10 additional users in Active Directory “Users” group
 - User1 .. User10
- Display all Active Directory users
 - Display a list of users in the “Domain Admins” group
- On DC1 display a list of all files in windows\system32 folder
 - Display only files that are created in last 10 days
 - Display only files that have been modified in last 10 days

Lab on working with certs and psdrives

- On DC1
 - Display a directory listing of all items in the CERT drive including subfolders
 - Display the members of the objects produced in the previous command
 - Identify a method from the member list that will retrieve the key algorithm for a certificate
 - Display a list of key algorithms for each installed certificate
 - Using Select-Object instead of ForEach-Object display a list of all certificates on the computer- display only columns “KeyAlgorithm” and “Issuer” .
The Key algorithm column should be a calculated column (using @{n=.. ; e= ...})
 - Create numeric objects 1 .. 100
 - Use the created objects as seed for random generation (use piping)
 - Discover the method to restart the computer from Get-WmiObject -Class Win32_OperatingSystem -EnableAllPrivileges